Dataset Identified

Bangladesh Demographic and Health Survey (BDHS) 2022 and 2017-18

Source, description, and exploratory data analysis

Predicting Maternal and Child Health Risk · a five-outcome risk cascade

1. Dataset at a glance

Field	Detail
Dataset name	Bangladesh Demographic and Health Survey (BDHS) — 2022 round (DHS-8) and 2017-18 round (DHS-7)
Producer	National Institute of Population Research and Training (NIPORT), Ministry of Health and Family Welfare, Government of Bangladesh, with ICF under The DHS Program (USAID-funded)
Recodes used	Individual Recode (IR) — one record per ever-married woman aged 15–49; Birth Recode (BR) — one record per birth or pregnancy
Native format	Stata .DTA with embedded value labels (also distributed as SPSS .SAV and flat ASCII); merged and cleaned to Apache Parquet, with a CSV mirror per cohort
Prediction targets	5 binary outcomes — facility delivery, neonatal death, stunting, severe stunting and child underweight — all measured in both survey rounds
Raw scale	2,134 variables before selection; 899 retained in the pooled master table
Pooled size	121,067 records × 899 columns (19 MB Parquet) — 73,239 from 2022 and 47,828 from 2017-18
Unit coverage	45,458 distinct mothers, 1,346 sample clusters, 38 strata across the two rounds
Sampling design	Stratified two-stage cluster sample: enumeration areas selected within strata, then households within clusters. Valid inference requires the sampling weight (v005), the primary sampling unit (v021) and the stratum (v022).
Access	Free on registration; a short research-purpose statement is approved by The DHS Program before download. Redistribution of the raw files is not permitted, so this project ships code and derived artefacts rather than source data.
Licence / ethics	Secondary analysis of de-identified public-use microdata. No personal identifiers are present; no additional ethical clearance was required.

2. Source links

The data are obtained from The DHS Program, which distributes the Bangladesh survey rounds on request. The three links below are the catalogue used for download, the report used to validate the prepared data, and the recode documentation used to decode the variables.

- **Dataset catalogue (download point for both rounds):** https://dhsprogram.com/data/available-datasets.cfm
- **BDHS 2022 Key Indicators Report (validation reference):** https://dhsprogram.com/pubs/pdf/PR148/PR148.pdf

- **Survey documentation and recode manuals:** https://dhsprogram.com/publications/publication-dhsg4-dhs-questionnaires-and-manuals.cfm

The DHS Program blocks automated requests, so these addresses must be opened in a browser after signing in to a registered account.

3. Short description

The BDHS is the nationally representative household survey of reproductive-age women in Bangladesh, run roughly every four years by NIPORT with technical support from ICF under The DHS Program. Each round interviews ever-married women aged 15–49 about their full reproductive history, the care they received during each recent pregnancy and delivery, the health and vaccination status of their young children, and the composition, assets and sanitation of their household. A subsample of women and children is physically measured for height and weight.

This project pools two rounds — BDHS 2022 (DHS-8) and BDHS 2017-18 (DHS-7) — merged from the Individual and Birth recodes and cleaned by one identical pipeline, giving a master table of 121,067 records × 899 columns. Pooling raises the training sample by roughly 65%, which matters most for the rarest outcomes, and it allows a trend comparison in which a difference between years cannot be an artefact of differing preparation. The purpose of the dataset in this project is to predict five binary maternal and child health outcomes from information a community health worker could collect at a first antenatal contact.

3.1 The two rounds, and where they differ

The rounds are not interchangeable. DHS-8 introduced a full pregnancy history, so 2022 records miscarriage, abortion and stillbirth; DHS-7 has only a live-birth history. Two harmonisations were required: Bangladesh renamed two divisions between rounds (Chittagong → Chattogram, Barisal → Barishal), and cluster and mother identifiers are namespaced by survey, because cluster 1 in 2017-18 is a different village from cluster 1 in 2022. Without that namespacing the two villages merge into one and every confidence interval in the analysis is understated.

Attribute	BDHS 2022	BDHS 2017-18
DHS phase	DHS-8	DHS-7
Records contributed	73,239 pregnancies	47,828 births
Mothers	27,324	18,134
Sample clusters	674	672
Strata	16	22
Reproductive history	Full pregnancy history (p* variables) — records miscarriage, abortion and stillbirth	Birth history only — live births alone
Vaccination module	Empty in the released recode	Complete (h2–h9)
Size at birth (m18)	Present	Column present but empty

This asymmetry decided the panel. Every outcome that exists in only one round — stillbirth, pregnancy loss and small size at birth (2022), incomplete and no immunization (2017-18) — was dropped, so none of the five modelled targets is single-round. All five are measured in both 2022 and 2017-18, which means every model trains on the pooled sample and the second round strengthens the evidence rather than fragmenting it.

3.2 Why one flat table is not possible: nested universes

DHS is a set of nested universes. Every woman answers the household and demographic modules; only women with a birth in the last five years answer the maternity module; only a subsample of their children is physically measured. A column therefore appears to be 93% missing simply because it belongs to an inner ring — it was never asked of the outer records. Any global missingness threshold deletes the inner rings first, which is precisely where the maternal-care and child-health outcomes live.

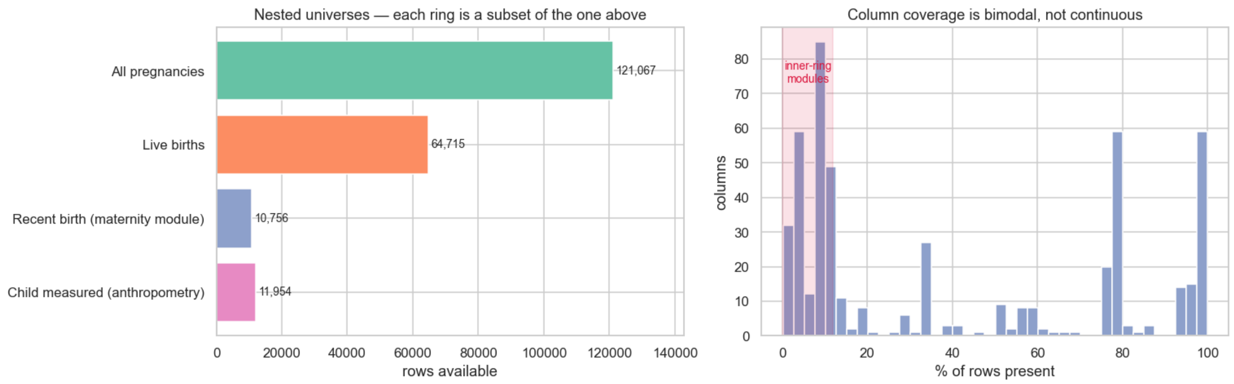

Figure 1 — Left: the nested universes, each a subset of the one above. Right: column coverage is bimodal — columns cluster either near 100% (asked of everyone) or under 12% (asked of one inner ring), with no natural cut-point in between. A "≥70% missing" rule does not separate good columns from bad ones; it separates outer rings from inner rings.

The pooled master is therefore sliced into four cohorts by universe membership, never by dropping rows with missing features. Only inside a cohort does a missingness figure mean anything, and a 30% within-cohort coverage floor is applied there.

Cohort	Records	Cols	Mothers	Rows/mother	Targets	Universe
cohort_births	112,546	128	45,067	2.50	1	All live births
cohort_mother	45,458	126	45,458	1.00	0	One record per woman — features and EDA only
cohort_measured_child	11,954	130	10,667	1.12	3	Children physically measured
cohort_recent_birth	10,773	142	10,156	1.06	1	Births in the last five years

Three further cohorts — pregnancy (73,239 records), child illness (16,883) and immunization (5,117) — were built for an earlier fifteen-outcome panel. Every target they carried was dropped, so they are no longer produced. All four surviving cohorts span both survey rounds; 124 columns form a shared feature core common to all of them, which is what makes one woman's features portable across all five models.

Rows per mother above 1 is the reason every split in the modelling notebook is grouped on `mother_id`. A plain random split would put siblings on both sides and inflate every score. Note also that `caseid` carries 45,463 unique values against `mother_id`'s 45,458 — five women appear under two fixed-width spellings each, so the raw string is not a safe grouping key.

3.3 The five prediction targets

Targets were chosen to be clinically actionable, measurable from the recodes in both rounds, and ordered so the panel tells a connected story: where she delivers → whether the baby survives → how the child grows.

Phase	Target	Cohort	Positive case	Prevalence
Delivery	facility_delivery	recent_birth	delivered in a health facility	57.0%
Newborn	neonatal_death	births	died within 28 days of birth	4.0%

Child	stunted	measured_child	height-for-age below −2 SD	28.1%
Child	severe_stunting	measured_child	height-for-age below −3 SD	7.8%
Child	underweight_child	measured_child	weight-for-age below −2 SD	21.8%

The panel was cut down from an earlier fifteen, and the reason each outcome was removed is worth recording. Nine could not be predicted from what a household survey collects — child fever reached ROC-AUC 0.573 at recall 0.004, wasting 0.588 at 0.005 — and a tenth, incomplete immunization, was dropped precisely because it scored well: at 0.944 it was predicting the child's age, since every child under six months is “incomplete” before the vaccination schedule finishes. A high score measuring the wrong thing is more dangerous than a low one.

The dropped outcomes are still derived and still stored as columns, so the descriptive sections below can report them. They are simply not modelled.

C-section is retained as a descriptive comparison but was dropped as a prediction target: it is a clinician's decision rather than a patient risk, so predicting it months in advance has little operational use. Its wealth gradient, however, is the sharpest equity finding in the survey and is reported in section 4.8.

4. Exploratory data analysis

All estimates below are survey-weighted and carry design-based confidence intervals computed by Taylor linearisation, which measures variance between clusters within strata rather than between individuals. The mean design effect across the outcomes examined is roughly 2.2, meaning a naive analysis that ignored clustering would report intervals about 1.5 times too narrow. Estimates quoted as national figures are computed per round: a pooled prevalence is a weighted blend of two populations and matches no published figure, because it is not one.

All figures in this section were produced by 01b_exploratory_analysis.ipynb and 01_data_preparation.ipynb and are reproduced here directly from those notebooks' executed output.

4.1 Structure, types and cardinality

View	Result
Shape	121,067 rows × 901 columns (899 stored + 2 design keys derived on load); 742 MB in memory with deep accounting
Dtype mix	528 category, 297 float64, 74 object, 2 string — roughly 540 categorical columns against ~170 numeric
Roles	single_round 361 · categorical 355 · numeric 116 · target 27 · metadata 19 · derived 11 · numeric_composite 9 · empty 1
Cardinality	binary 410 · few (3–10) 227 · many (11–50) 161 · high (>50) 74 · constant 28 · empty 1
Dead columns	29 columns are constant or entirely empty and carry zero information; they are filtered downstream by a nunique() > 1 check

The dtype ratio is the single fact that shapes all preprocessing. DHS is overwhelmingly a categorical survey, so one-hot encoding is the dominant cost in every pipeline, and 410 of 901 columns being binary is what keeps that cost manageable. Reading mean coverage by role gives the numeric signature of the nested structure: metadata averages 93% present because identifiers and weights exist on every row, while targets average 27% because each belongs to an inner module.

4.2 Missingness structure

Variables in DHS do not go missing independently — they go missing in blocks, because an entire module was out of universe. The Jaccard matrix makes those blocks visible: a value of 1.0 means two variables are always missing on exactly the same rows.

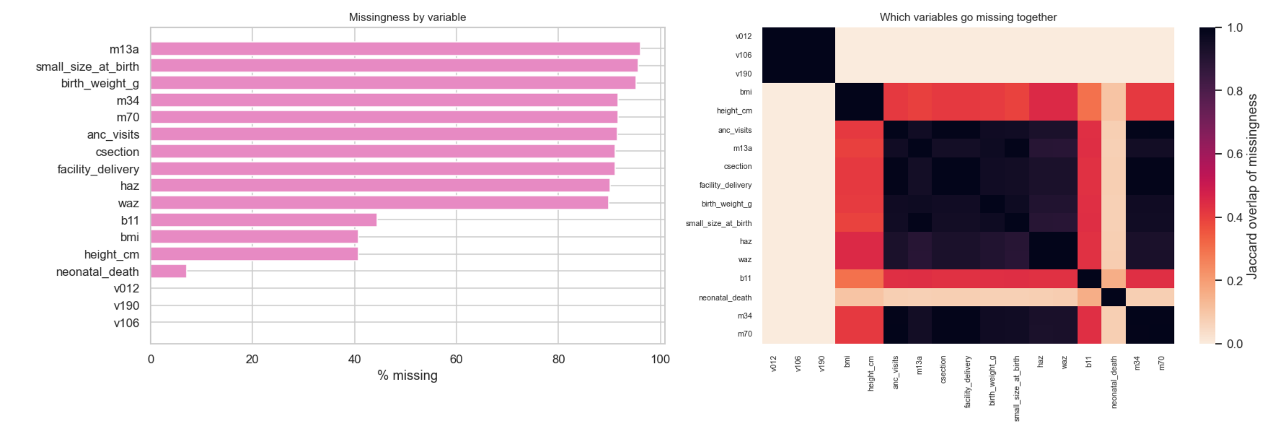

Figure 2 — Left: missingness by variable. Right: which variables go missing together. The bright blocks are survey modules, not data-quality problems — the maternity module (csection, facility_delivery, anc_visits, m34, m70) is asked only of recent births, and anthropometry (haz, waz) only of children physically measured.

4.3 Cohort composition and nesting

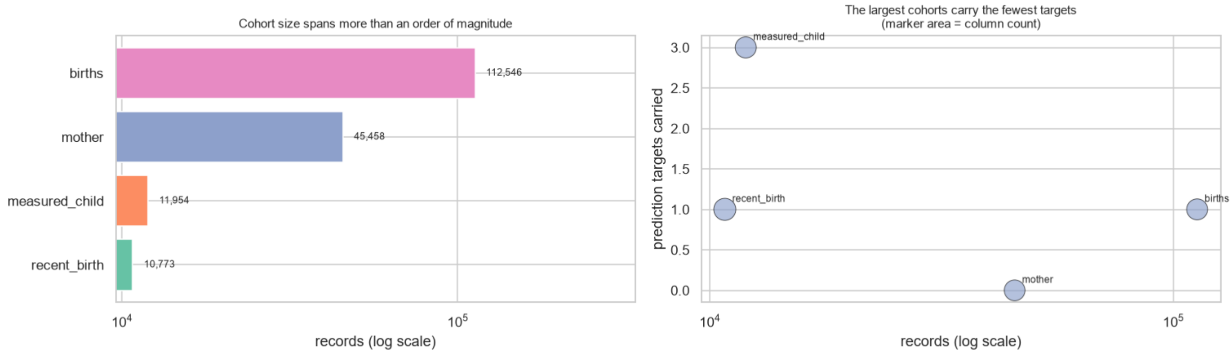

Figure 3 — Cohort size spans more than an order of magnitude, and the largest cohort carries the fewest targets. cohort_births holds 93% of the master and supports one outcome; cohort_measured_child holds a tenth as many records and supports three. Marker area is column count.

The relationship is a direct consequence of the nesting: the questions that produce interesting outcomes — anthropometry, vaccination cards, antenatal care — are asked of progressively narrower groups, so breadth of measurement is bought with sample size. This is why a single model-selection recipe cannot be applied across the panel. An outcome resting on 112,546 births tolerates a complex model and yields tight intervals; one resting on 11,954 measured children does not.

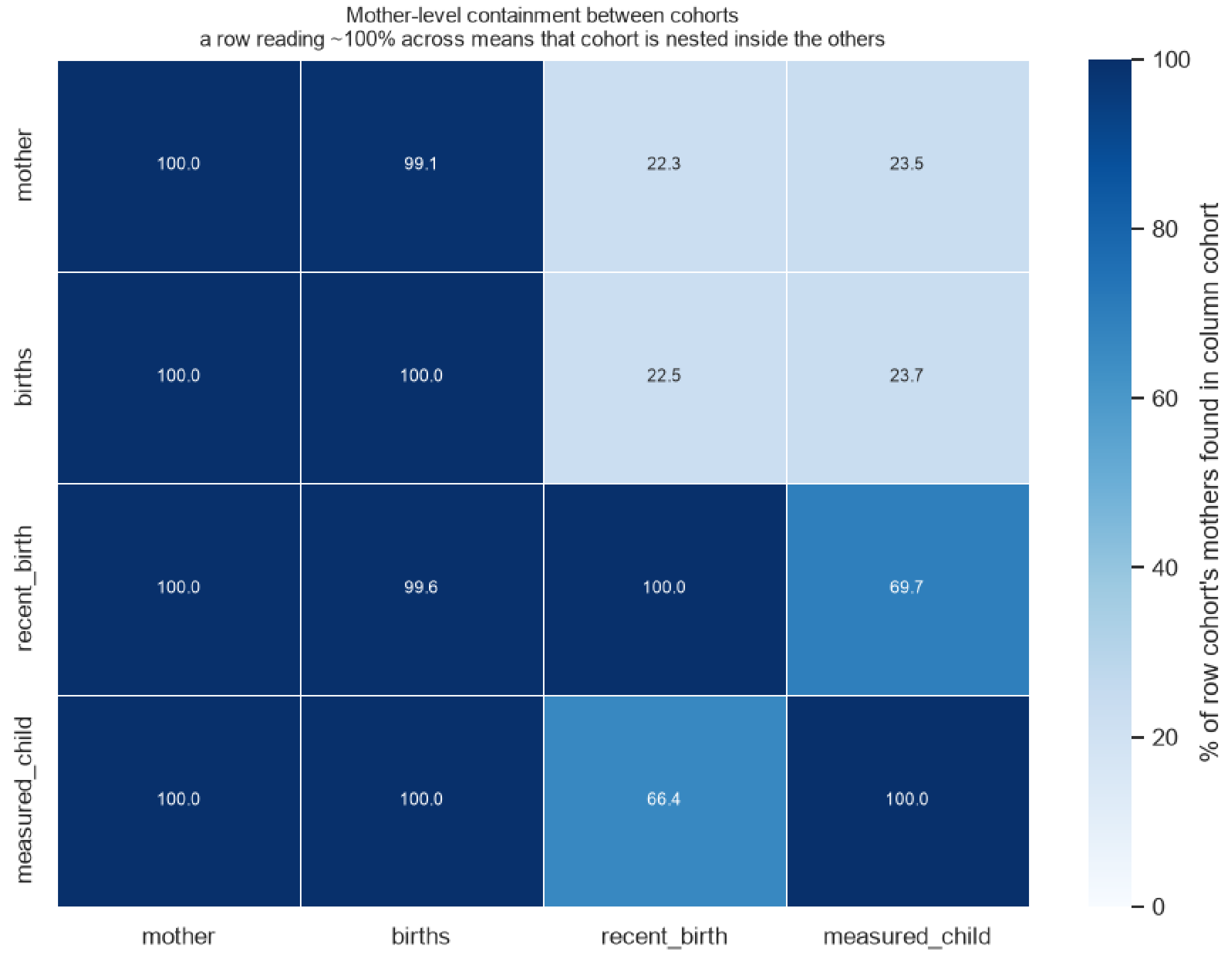

Figure 4 — The nesting claim, tested rather than assumed. Each cell reads as "this share of the row cohort's mothers also appear in the column cohort". Sub-universes come out near 1.0 against their parents, confirming the cohort definitions are genuine subsets.

4.4 Weighted national characteristics

Computed on 45,458 mothers, one record per woman, weighted to national representation with a design-based confidence interval on every row. The five wealth quintiles each land near 20%, which they must by construction — if they did not, the weighting would be broken — and residence comes out at 71.6% rural / 28.4% urban, matching the national split.

Characteristic	Level	Weighted %	95% CI	n
Age group	15–19	5.1	[4.8, 5.3]	2,178
	20–24	15.6	[15.2, 16.0]	6,976
	25–29	18.3	[17.9, 18.8]	8,330
	30–34	18.4	[18.0, 18.8]	8,344
	35–39	17.2	[16.9, 17.6]	7,930
	40–44	13.3	[12.9, 13.6]	6,095
	45–49	12.0	[11.7, 12.3]	5,605
Residence	rural	71.6	[70.8, 72.4]	29,249
	urban	28.4	[27.6, 29.2]	16,209

The gap between the n column and the weighted percentage is the reweighting made visible. Sylhet contributes roughly 4,900 sampled women but represents only 5.8% of the population: DHS deliberately oversamples smaller divisions so each has enough cases to analyse on its own, which is exactly why unweighted percentages would misdescribe the country.

Variable	n	Weighted mean	SD	Median	IQR	Min	Max	Missing %
Age (years)	45,458	32.72	8.61	33.00	26.0–40.0	15.0	49.0	0.0
Years of education	45,458	6.05	4.23	6.00	3.0–9.0	0.0	18.0	0.0
Age at first marriage	36,312	16.17	3.26	16.00	14.0–18.0	10.0	49.0	20.1
Age at first birth	45,067	18.46	3.53	18.00	16.0–20.0	10.0	47.0	0.9
Children ever born	45,458	2.47	1.39	2.00	1.0–3.0	0.0	13.0	0.0
BMI (kg/m^2)	26,939	23.61	4.28	23.27	20.5–26.3	12.2	59.5	40.7
Height (cm)	26,940	151.01	5.53	151.00	147.4–154.6	100.7	198.0	40.7
Weight (kg)	26,945	53.94	10.66	53.00	46.2–60.6	24.0	148.3	40.7
Composite risk flags	45,458	0.86	0.92	1.00	0.0–1.0	0.0	5.0	0.0

The Missing % column is not random. Height, weight and BMI are absent for about 41% of women because anthropometry was measured on a subsample, not because anyone refused. Age and education are complete because they are asked of everyone.

The min and max columns double as an implausibility check: BMI inside roughly 12–48, height inside 100–200 cm, birth weight inside 500–6,000 g and age inside 15–49. Anything outside those bounds would mean a DHS special code survived the decoding pass. Nothing does.

4.5 Univariate distributions

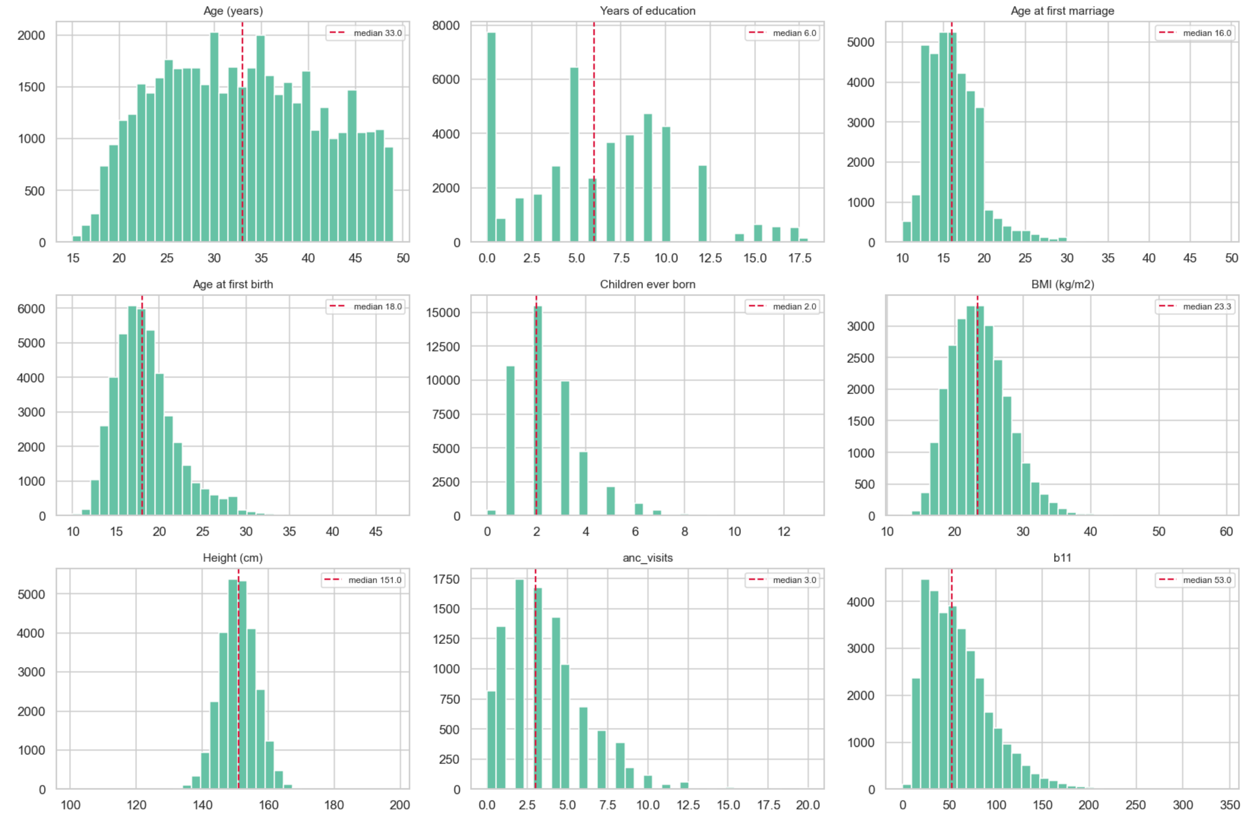

Figure 5 — Nine continuous distributions with the median marked. Age at first marriage and age at first birth both pile up in the late teens; years of education is multi-modal, spiking at 0, ~5 and ~10 — the points where schooling stops; BMI is right-skewed, so its mean sits above its median.

Two features of this panel drive later decisions. Antenatal visits has a visible spike at zero — 426 women received no antenatal care at all. They survive in this dataset only because the cleaning step read the label “no antenatal visits” as a semantic zero rather than as missing; an automatic numeric coercion would have deleted the highest-risk group in the survey. Birth interval has a long right tail, so its mean overstates typical spacing and the median is the honest summary.

Figure 6 — Eighteen categorical characteristics across five themes, weighted to national shares. Bar height is share of the population, not of the sample.

Read as five connected accounts rather than eighteen charts. Wealth quintiles sit near 20% each, as they must. Setting shows the sampling correction most clearly — Dhaka is largest at 25%, while Sylhet and Barishal sit near 6% each despite being deliberately oversampled in the raw counts. Health insurance is essentially absent, so almost all care is paid out of pocket, which is important context for the facility delivery wealth gap reported below: where care is bought rather than covered, access tracks income directly. Distance to a facility is reported as a big problem by 44% of women — the closest thing in the survey to a supply-side variable.

4.6 Outcome prevalence with design-based intervals

Outcome	Cohort	n	Prevalence %	95% CI	DEFF
facility_delivery	recent_birth	10,773	57.04	55.42 – 58.66	3.01
neonatal_death	births	112,546	4.03	3.87 – 4.18	1.82
stunted	measured_child	11,954	28.13	27.04 – 29.22	1.83
severe_stunting	measured_child	11,954	7.75	7.08 – 8.42	1.93
underweight_child	measured_child	11,950	21.83	20.85 – 22.82	1.75
infant_death *	births	112,546	5.46	5.28 – 5.64	1.80
under5_death *	births	112,546	6.43	6.24 – 6.63	1.82

mother_underweight *	mother	26,939	10.47	10.00 – 10.93	1.60
mother_overweight *	mother	26,939	34.72	33.86 – 35.59	2.31
low_birth_weight *	recent_birth	5,784	15.45	14.28 – 16.61	1.56
csection *	recent_birth	10,756	38.28	36.80 – 39.77	2.63

** descriptive outcomes, not part of the five-target panel. All values are pooled across rounds and are the correct input for modelling, but the wrong thing to quote as national figures — see section 5.*

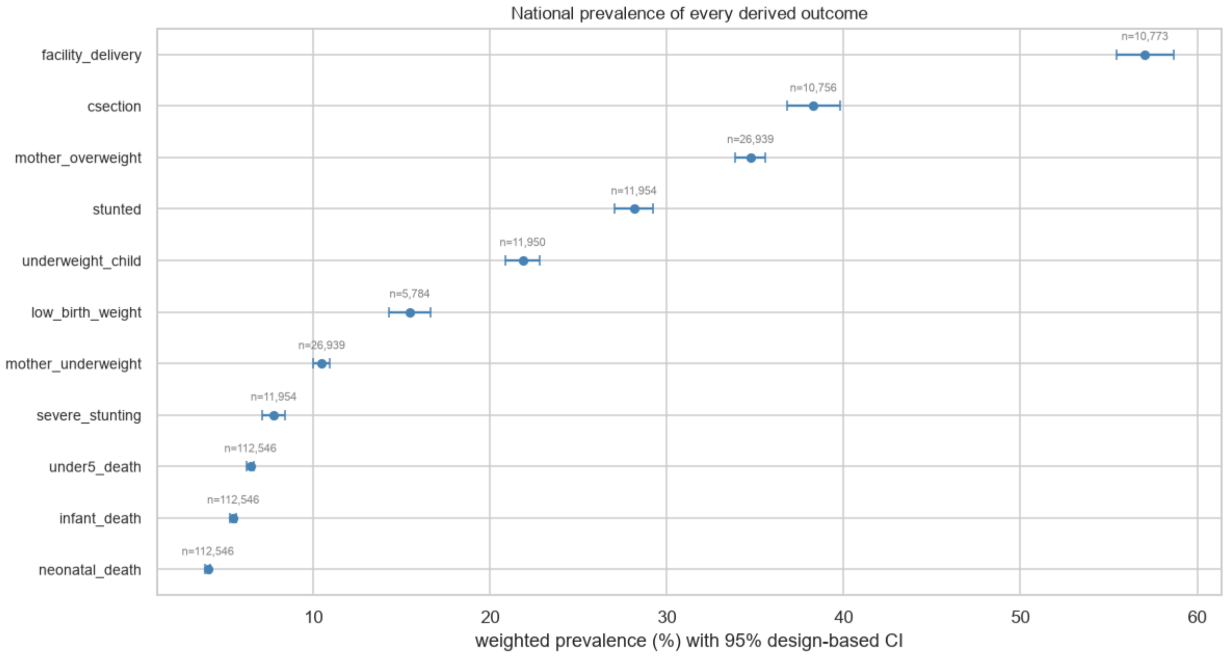

Figure 7 — The same estimates as a forest plot, sorted by prevalence and annotated with sample size. The visual point is the width of each interval rather than its position: rare outcomes have intervals that look narrow in absolute percentage points but are wide relative to the estimate itself.

Read n alongside the estimate. `neonatal_death` rests on 112,546 births and is tight at 4.03% ± 0.16; `facility_delivery` rests on 10,773 and is correspondingly wider at 57.04% ± 1.6 — and it carries the highest design effect in the panel at 3.01, because delivering in a facility is strongly clustered by village. An outcome measured on 10,000 records cannot support the same claims as one measured on 112,000, and the interval width says so directly. No target in the panel is single-round: all five are measured in both survey rounds.

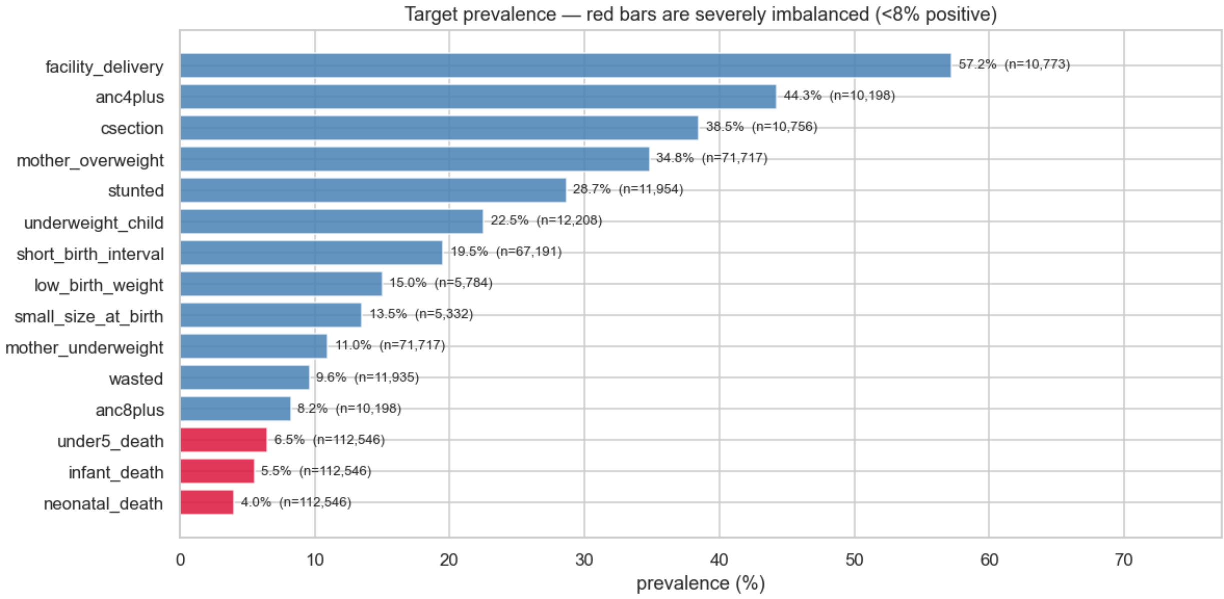

Figure 8 — Prevalence across all derived outcomes, not only the five modelled ones; red bars mark severely imbalanced outcomes below 8% positive. The panel's own five are facility delivery, neonatal death, stunting, severe stunting and child underweight.

4.7 Class balance

Two numbers matter here and they are not the same. The imbalance ratio decides which metrics are meaningful; the absolute count of positives decides whether the outcome is learnable at all. No reweighting recovers information that was never in the sample.

Target	n	Positives	Positive %	Imbalance	Regime
facility_delivery	10,773	6,167	57.2	0.7 : 1	balanced
stunted	11,954	3,427	28.7	2.5 : 1	moderate
underweight_child	12,208	2,752	22.5	3.4 : 1	moderate
severe_stunting	11,954	948	7.9	11.6 : 1	severe
neonatal_death	112,546	4,535	4.0	23.8 : 1	severe

Imbalance spans 0.7 : 1 to 23.8 : 1 — a thirty-fold range across five outcomes. Three regimes need three treatments. facility_delivery is genuinely balanced, so accuracy means something. For neonatal_death at 23.8 : 1, a model predicting “survives” for every record is 96.0% accurate and clinically worthless: the metric has to change, not just the model.

But the ratio is only half the story. `neonatal_death` has the worst ratio yet holds 4,535 positive cases drawn from 112,546 births — a difficult problem with enough signal to attempt, and the reason a rare outcome stayed in the panel while several commoner ones did not. `severe_stunting` looks easier at 11.6 : 1 but rests on 948 positives in total, which makes it the scarcest target in the panel; a five-fold split leaves roughly 190 cases per fold to validate against. This is why the modelling notebook selects on PR-AUC rather than accuracy and reports bootstrap confidence intervals throughout.

4.8 Equity gradients

Before modelling, the classic DHS social gradients were confirmed to be present. If wealth and education did not predict these outcomes, something would be wrong with the decoding.

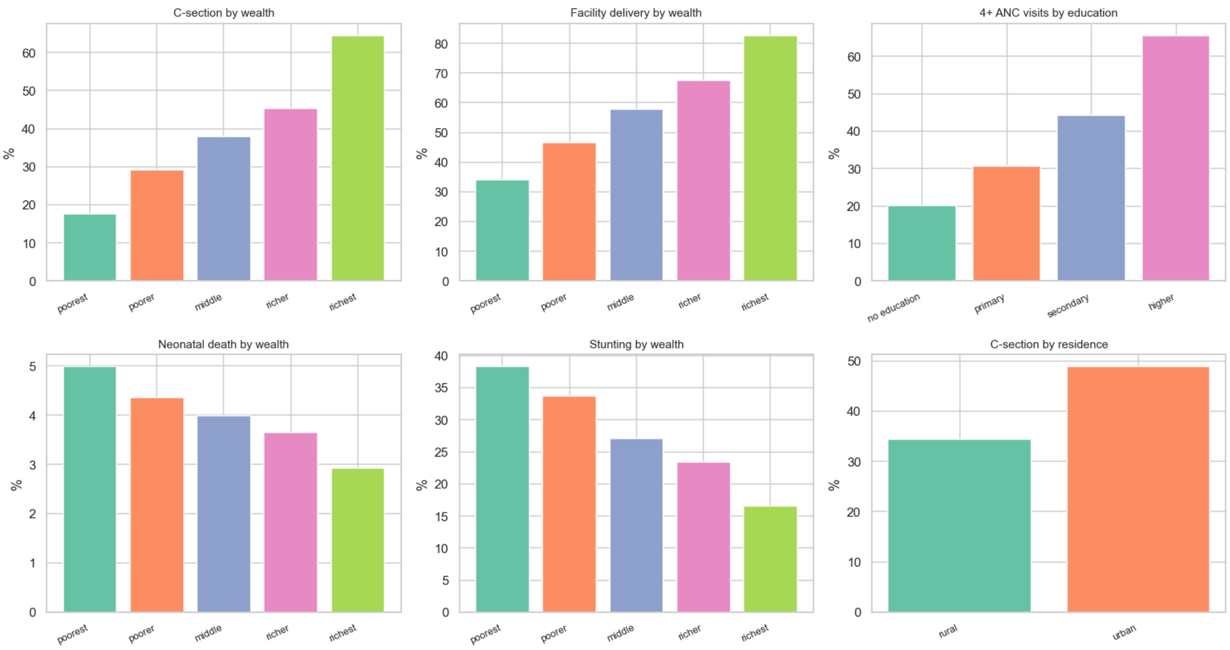

Figure 9 — Social gradients across wealth quintile, education and residence, all survey-weighted. The C-section gradient is steep and monotonic; the neonatal-death gradient is much flatter.

Across the six outcomes examined (the five targets plus C-section as a descriptive comparison), the poorest quintile carries the disadvantage on every one. That consistency — across service access, nutrition and mortality alike, in different directions and on wildly different scales — is the finding, not any single row.

Outcome	Poorest %	Richest %	Gap (pp)	Ratio	Worse off
facility_delivery	34.0	82.6	+48.6	2.43	poorest
csection *	17.6	64.4	+46.8	3.65	poorest
stunted	38.3	16.5	−21.7	0.43	poorest
underweight_child	29.9	12.9	−16.9	0.43	poorest
severe_stunting	11.9	3.6	−8.3	0.30	poorest
neonatal_death	5.0	2.9	−2.1	0.59	poorest

** descriptive, not a modelled target.*

- **Access moves by tens of points.** Facility delivery runs from 34.0% in the poorest quintile to 82.6% in the richest — a 48.6-point gap — and C-section 46.8 points, from 17.6% to 64.4%. Access is bought, and in a system with almost no health insurance it is bought with cash.
- **Chronic undernutrition tracks wealth closely.** Stunting falls 21.7 points across the distribution and child underweight 16.9. Both accumulate over years of household conditions, which is why they follow the wealth gradient so faithfully.
- **Survival moves by two points.** Neonatal death runs 5.0% in the poorest quintile against 2.9% in the richest — a 2.1-point gap, an order of magnitude smaller than the access gaps. Mortality is driven more by biological and care-timing factors than by wealth alone, which is precisely what makes it worth modelling rather than merely tabulating.
- **The direction of harm depends on the outcome's kind.** A positive gap on a pathway outcome such as facility delivery means the poor are underserved; a negative gap on an adverse outcome such as stunting is the same finding wearing the opposite sign. The “worse off” column resolves that, which is why it is carried in the table rather than left to the reader.

The C-section row deserves the hardest look. Its wealth gradient does not plausibly reflect obstetric need — medical indications for caesarean do not track household wealth. Under-service at the bottom is certainly part of it, but 64.4% at the top is far above what need would justify. Both ends are wrong, in opposite directions, and no single-signed measure can express that. It is the most policy-relevant descriptive finding in this analysis, and the reason C-section was kept as a description after being dropped as a prediction target.

4.9 Geography — the eight divisions

Division	n	C-section %	Facility delivery %	ANC 4+ %	Small size at birth %	Neonatal death %
khulna	1,152	54.8	72.3	49.5	11.3	4.00
dhaka	1,590	47.8	64.1	49.3	16.1	3.82
rajshahi	1,093	44.0	61.6	42.6	11.1	4.63
rangpur	1,213	34.9	55.0	46.6	12.8	4.64
mymensingh	1,316	31.8	46.2	45.0	11.0	4.73
barishal	1,144	31.2	46.3	32.3	12.5	3.40
chattogram	1,843	28.2	52.4	37.3	15.8	3.21
sylhet	1,422	24.3	44.5	32.3	17.8	4.73

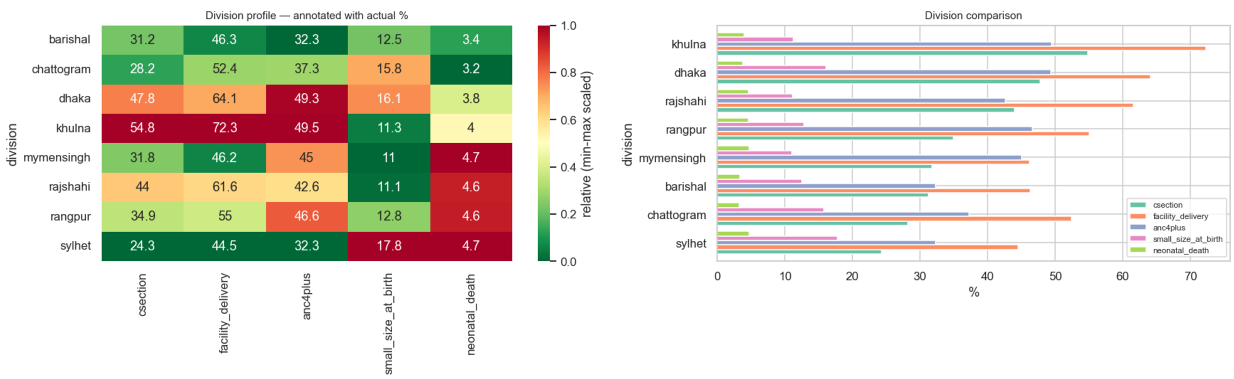

Figure 10 — Division profile as a heatmap and as grouped bars. Colour is min-max scaled within each column, so a dark cell means "worst among the eight divisions" rather than "bad in absolute terms" — always read the printed percentage alongside the colour.

The question to ask is whether a division lags on everything or whether divisions trade off. Sylhet is last on C-section, facility delivery and ANC, has the highest share of small babies and is tied for the highest neonatal mortality — a consistent service-availability problem. Barishal reads differently: it is near the bottom on facility delivery and ANC but has the second-lowest neonatal mortality, which points at a referral and access pattern rather than a uniform failure of care.

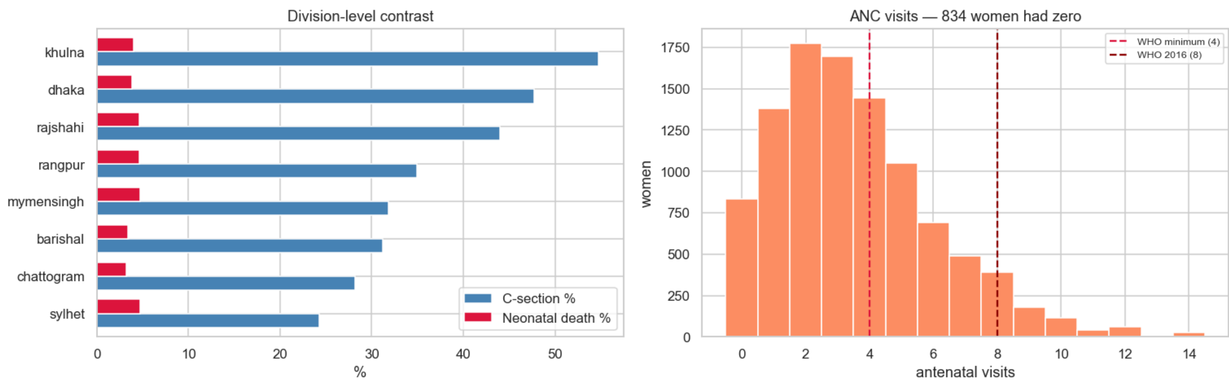

Figure 11 — Left: division-level contrast between C-section and neonatal death. Right: the distribution of antenatal visits against the WHO minimum of four and the 2016 recommendation of eight — 426 women received none at all.

4.10 Correlation and multicollinearity

Correlation is computed per cohort, never on the pooled master: the master stacks seven universes, so two columns can appear uncorrelated there purely because they are non-missing on disjoint sets of rows. Every cohort carries between 2 and 12 pairs above |r| = 0.85 out of 400 to 1,600 pairs — consistently around 1% or less, which is the expected result for a feature core that was assembled deliberately rather than scraped. More features does not mean more redundancy: recent_birth has the most numeric columns (40) and one of the lowest redundant fractions (0.50%), while immunization has 34 and the highest (1.04%).

Three families recur across cohorts because they come from the shared core:

- **The fertility-accounting block** — v201 (children ever born), v218 (living children), v219 (living plus current pregnancy) and v220 (living, capped) reach r = 0.996 and are arithmetically related. This is the same family that produced leakage on neonatal_death.
- **The age block** — v012 with v512 (years since first marriage) at 0.935, and v511 (age at marriage) with v525 (age at first sex) at 0.964. In this population marriage, first sex and first birth follow each other closely.
- **The anthropometry pair** — bmi and weight_kg at 0.927 in all four cohorts. BMI is weight over height squared, and height varies far less than weight, which is why the correlation is this high rather than moderate.

Measured across the four surviving cohorts, these families are strikingly stable: the fertility pair v201 × v218 runs 0.917–0.957, the age pair v012 × v512 runs 0.847–0.935, v511 × v525 runs 0.830–0.913, and bmi × weight_kg sits at 0.926–0.927 in every one. Redundancy here is a property of the shared feature core rather than of any single cohort, which is why it should be resolved once and globally rather than per target.

The two exactly-redundant pairs previously flagged — full_immunization × incomplete_immunization, a definitional complement, and pregnancy_loss × stillbirth, which differed only in universe — both sat in cohorts that are no longer built. Neither survives into the five-target panel.

No global pruning is applied on the basis of these figures, because what to do about redundancy depends entirely on the model. Random Forest, XGBoost and LightGBM are unaffected — a tree simply picks one of a correlated pair at each split. Logistic regression, SVM and KNN are not: coefficients become uninterpretable and distance metrics double-count whatever the pair measures. The analysis is therefore diagnostic — it says which coefficients to distrust, and it explains why a tree's feature importance can look unstable when two of its inputs are near-duplicates.

4.11 Data quality

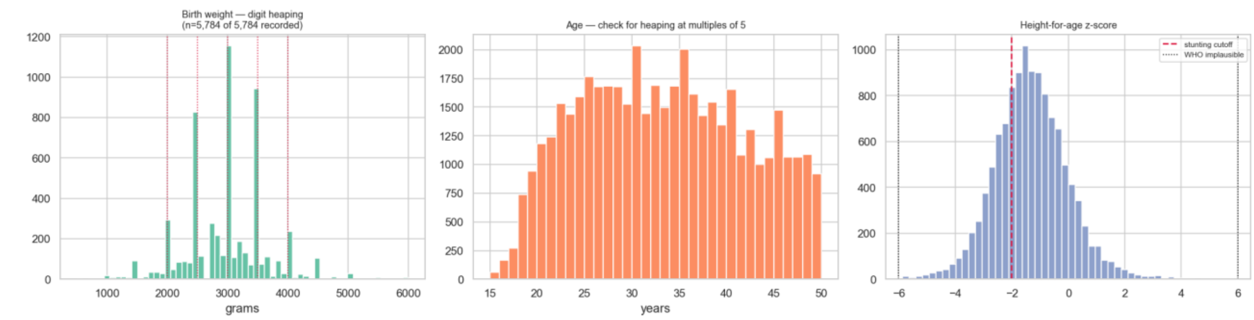

Figure 12 — Three data-quality checks: digit heaping in recalled birth weight, age heaping at multiples of five, and height-for-age z-scores against the WHO plausible range.

Check	Result	Reading
Birth weight on a round 100 g	96.5%	Recalled, not recorded
Birth weight on a round 500 g	63.2%	Mothers reporting “three kilos” rather than reading a card

Birth weight reported at all	5,784 of 10,756 recent births (53.8%)	Nearly half of recent births carry no weight
HAZ outside WHO ±6 SD	0 of 11,954	The “flagged cases” label was correctly nulled during cleaning
Size at birth (m18) available	5,332 — 2022 only	Column present but empty in the 2017-18 recode
Women with zero antenatal visits	426	Preserved as a semantic zero, not deleted as missing

Birth-weight heaping is the consequential one. The low-birth-weight definition is a hard cut at exactly 2,500 g, and reported values pile up precisely on that threshold, so small differences in rounding flip babies across the boundary. This is the concrete reason small_size_at_birth is the primary newborn-size target and low_birth_weight only a secondary check.

5. Validation against published reports

Indicators were recomputed independently from the raw recodes and checked against the published DHS reports, per round. Validation is per round rather than pooled, because a pooled prevalence is a weighted blend of two populations and matches no published figure by construction.

Round	Indicator	This dataset %	Published %	Difference
BD 2022	Stunting (under-5)	23.22	23.6	−0.38
	Wasting (under-5)	11.04	11.0	+0.04
	C-section	43.59	45.5	−1.91
	Facility delivery	63.84	65.3	−1.46
BD 2017-18	Stunting (under-5)	30.72	31.0	−0.28
	Wasting (under-5)	8.34	8.4	−0.06
	Underweight (under-5)	21.67	22.0	−0.33
	C-section	32.77	33.0	−0.23
	Facility delivery	49.97	49.6	+0.37
	ANC 4+ visits	47.01	47.0	+0.01

Every indicator lands within four percentage points, the largest passing deviation being 1.91. The cleaning, decoding, weighting and variance estimation would all have to be wrong in the same direction, in two separate surveys, to fake that.

One check fails, and it is a real finding. `child_ari` — an outcome since dropped from the panel — came out around four times the published rate. Fever and diarrhoea — derived by the same mechanism from the same module, in the same cell of the cleaning script — match almost exactly, which locates the fault precisely: not in the pipeline, but in the definition of this one target. DHS defines ARI symptoms as cough plus short rapid breathing that was chest-related; the derivation here used only the short-rapid-breathing question and dropped the chest-related qualifier. Applying the full definition reproduces the published figure (2.9% against 3.0% for 2017-18). A target can pass every internal reconciliation and still measure something other than what its name claims.

Two of the five targets have no published counterpart on a comparable denominator — neonatal mortality is published as a rate per 1,000 births over a five-year window rather than as a proportion of records, and severe stunting is reported inside the stunting distribution rather than as a separate binary. Those are recorded as not validated rather than quietly omitted, so that “not validated” is never mistaken for “validated and passing”.

6. Summary of findings

1. **Missingness is structural, not a quality problem.** Column coverage is bimodal — near 100% or under 12% — with no natural cut-point between. The Jaccard matrix confirms variables go missing in module-shaped blocks. This is the justification for the cohort design and against any global missingness threshold.
2. **The panel is severely and unevenly imbalanced.** Positive rates span 57.2% (facility delivery) to 4.0% (neonatal death), an imbalance range of roughly thirty-fold. Absolute positive counts matter as much as ratios: severe_stunting holds only 948 positives in total.
3. **Design effects are substantive.** Mean DEFF is around 2.2, so an analysis ignoring clustering would report confidence intervals roughly 1.5 times too narrow. Every interval in this document is design-based.
4. **Wealth stratifies access far more than it stratifies biology.** Facility delivery moves 48.6 percentage points across the wealth distribution and C-section 46.8, while neonatal death moves 2.1. The poorest quintile is worse off on all six outcomes examined.

5. **Geography shows both uniform lag and genuine trade-offs.** Sylhet lags on every service indicator and carries the highest share of small babies. Barishal is low on facility delivery yet low on neonatal mortality, which points at a referral pattern rather than a uniform failure of care.
6. **Birth weight is recalled rather than recorded.** 96.5% of reported weights fall on a round 100 g and 63.2% on a round 500 g, and only 53.8% of recent births carry any weight. This makes the 2,500 g cut-off unstable, and is why neither low_birth_weight nor small_size_at_birth survived into the modelled panel.
7. **The prepared data reproduce the published national figures.** Ten indicators across two rounds fall within four percentage points of their published values, the largest deviation being 1.91. The single failure, child_ari, was traced to a definitional omission rather than a pipeline fault; that outcome is no longer in the panel.

7. Caveats

- The survey is cross-sectional and retrospective. Everything reported here is an association, not a causal effect.
- Covariates describe the mother at the time of interview, not at the time of an older pregnancy.
- Pooled prevalences are the correct input for modelling but should not be quoted as national figures; use the per-round estimates in section 5.
- The Rao–Scott correction applied to the chi-square tests is the simple mean-DEFF approximation rather than the full second-order version. It is adequate for screening associations; a formal test should cite a proper survey package.
- DHS records no clinical pregnancy complications — no blood pressure, glucose, pre-eclampsia, gestational diabetes, haemorrhage or sepsis — and the anaemia biomarker recode was not merged into either round.
- The raw DHS files cannot be redistributed. This project ships code and derived artefacts; the source data must be requested from The DHS Program directly.

Sources for every number in this document

All tables and figures are taken from the executed output of `01_data_preparation.ipynb` and `01b_exploratory_analysis.ipynb`, which read `bdhs_clean.parquet` (121,067 × 899) and the seven cohort files in `cohorts/`. Figures are reproduced directly from those notebooks and are stored alongside this document in `doc/figures/`.